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AHEAD Online

Object Oriented Programming Using Java

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Objectives

- ✓ Java source file structure.
- ✓ Program compilation.

Structure of Java Program

```
class class-name {
        public static void main(String args[]) {
                statement1;
                statement2;
                    …
                    …
        }
}
```

Example Program

First.java

```
/* First Java Program */
class First {

        public static void main(String args[]) {
                System.out.println("Hello World");
        }

}
```

Example Program

Name of the source file - First.java

- ✓ The Java compiler requires that a source file with the .java filename extension.
- ✓ File name same as name of the class in which main resides [First].
- ✓ Java is case sensitive.

Compiling and Running

Compiling: is the process of translating source code written in a particular programming language into computer-readable machine code that can be executed.

```
$ javac First.java
```

This command will produce a file 'First.class', which is used for running the program with the command 'java'.

Running: is the process of executing program on a computer.

```
$ java First
```

Example Program

public static void main(String args[])

- ✓ main function where the program begins execution.
- ✓ public access specifier makes main function to be called from anywhere.
- ✓ JVM calls main before any object creation hence static keyword is used.
- ✓ String args[ ] declares a parameter named args, which is an array of instances of the class String.
- ✓ main function body should be with in '{'.

Example Program

```
System.out.println("Hello World.");
```

- ✓ Print the string Hello World.
- ✓ println( ) method displays the string which is passed to it with a newline.
- ✓ System is a predefined class that provides access to the system, and out is the output stream that is connected to the console.
- ✓ Java statements end with a semicolon.

Java IDEs

- ✓ Eclipse
- ✓ NetBeans
- ✓ BlueJ
- ✓ IntelliJ IDEA Community Edition

Takeaways

- ✓ Structure of java file.
- ✓ How to compile and run a java program?
- ✓ An Example program.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018*